

Simultaneous equations models



- 1 A cat moves down a path at 8 m/s. A dog begins moving down the same path from the same point at a speed of 12 m/s, but starts 2 seconds later.
 - a Graph the distance travelled by each animal, measuring the time from when the cat starts running.
 - b How long will it be before the dog catches up with the cat?
 - c How far from the initial point will this be?

- 2 A family owns two businesses, EcoLine and Helios. These businesses made a combined profit of \$12 million in the previous financial year, with Helios making 3 times as much profit as EcoLine.

Let x represent the profit (in millions of dollars) of EcoLine, and y be the profit (in millions of dollars) of Helios.

 - a Use the fact that the two businesses made a combined profit of \$12 million to set up an equation involving x and y .
 - b Use the fact that Helios made 3 times as much as EcoLine to set up another equation relating x and y .
 - c Graph the two equations on the same set of axes.
 - d Use the graph to find EcoLine's profit.
 - e Use the graph to find Helios' profit.

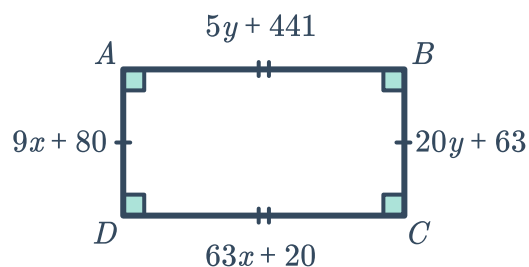
- 3 A rectangular zone is to be 2 cm longer than it is wide, with a total perimeter of 20 cm. Let y represent the length of the rectangle and x represent the width.
 - a Complete the following two equations that represent the information.
 - i $y = \text{ } + \text{ }$
 - ii $2x + \text{ } = 20$
 - b Graph the two equations on the same set of axes.
 - c Use the graph to find the length and width of the rectangle.
 - i Length
 - ii Width

- 4 A mother is currently 6 times her son's age. In 4 years time, she will be 4 times her son's age. Let x and y be the present ages of the son and mother respectively.
 - a Use the fact that the mother is currently 6 times her son's age to set up an equation relating x and y , where y is the subject of the equation.
 - b Use the fact that the mother will be 4 times her son's age in 4 years time to set up another equation relating x and y . Write the equation in the form $y = mx + b$.
 - c Solve the system of linear equations using technology.



- 5 Toby's piggy bank contains only 5c and 10c coins. He knows that there are 53 coins in the piggy bank, and that the total value is \$3.50. Let x and y be the number of 5c and 10c coins respectively. We will create two equations then solve them simultaneously to find the number of coins of each type.
- Use the fact that the total number of coins is 53 to set up an equation relating x and y .
 - Use the fact that the total amount of coins are worth \$3.50 to set up another equation relating x and y .
 - Solve the system of linear equations by either graphing or by using technology.
- 6 For two numbers x and y :
- Six times the first number is added to the second number to get 71.
 - The difference between eight times the first number and the second number is 83.
- Set up two equations relating x and y . Use x as the first number and y as the second.
 - Sum equation
 - Difference equation
 - Solve the system of linear equations in part (a), using appropriate technology.
- 7 There are 28 members in a monkey troupe, and the females outnumber the males by 4. Given this information, we want to find the number of each gender in the troupe. Let x be the number of male monkeys and y be the number of female monkeys in the troupe.
- Use the fact that the females outnumber the males by 4 to set up an equation with y as the subject.
 - Use the fact that there are a total of 28 members in the troupe to form another equation relating x and y . Write the equation in a form where the terms involving x and y are on one side, and the constant term is on the other.
 - Solve the system of linear equations using technology.
- 8 A gym offers aerobics classes where non-members pay \$3 per class and members pay a \$4 fee plus an additional \$2 per class. The monthly cost, y , of taking x classes can be modelled by the linear system:
- Non-members: $y = 3x$
 - Members: $y = 2x + 4$
- Graph the two equations on the same set of axes.
 - State the values of x and y which satisfy both equations.
 - What do the coordinates of the solution mean?

9 Consider the rectangle $ABCD$ given:



- a Use the fact that $AB = CD$ to set up Equation 1.
 - b Use the fact that $AD = BC$ to set up Equation 2.
 - c Solve the system of linear equations using technology.
 - d Find the length of the side CD .
 - e Find the length of the side AD .
- 10 The existing incandescent light bulbs in Buzz's home cost \$6.00 per month to operate. He is considering switching to new energy-efficient fluorescent bulbs, which will cost \$4.00 per month to operate. It will cost him \$4.00 to purchase the new bulbs.
- a Write an equation for the cost c of using the old incandescent bulbs for m months.
 - b Write an equation for the cost c of using the new energy-efficient fluorescent bulbs for m months.
 - c Graph the two equations on the same set of axes.
 - d Buzz decides to replace the old incandescent bulbs today and goes to the store to buy them. How many whole months from now will he start saving money overall?
- 11 Two twins, Amelia and Xavier, play soccer together. The twins have scored a combined total of 34 goals so far and Amelia has scored 8 more goals than Xavier. Let x be the number of goals scored by Amelia, and y be the number of goals scored by Xavier.
- a Use the fact that the twins have scored a combined total of 34 goals to set up an equation involving x and y .
 - b Use the fact that Amelia has scored 8 more goals than Xavier to set up another equation involving x and y .
 - c Graph the two equations on the same set of axes.
 - d Use the graph to find the number of goals Amelia has scored this season.
 - e Use the graph to find the number of goals Xavier has scored this season.

12 A clothing manufacturer is deciding whether to employ people or to purchase machinery to manufacture their line of t-shirts. After conducting some research, they discover that the cost of employing people to make the clothing is $y = 400 + 60x$, where y is the cost and x is the number of t-shirts to be made, while the cost of using machinery (which includes the cost of purchasing the machines) is $y = 1000 + 30x$.



- a Graph the two cost functions on the same set of axes.
- b Find the value of x at which it will cost the same whether the t-shirts are made by people or by machines.
- c Find the range of values of x for which it will be more cost efficient to use machines to manufacture the t-shirts.
- d Find the range of values of x at which it will be more cost efficient to employ people to manufacture the t-shirts.